

Name :- Preet B. Dhote

Reg. Np. :- 241070019

BLACK BOX TESTING (ECP + BVA + Functional)

```
B01 ✓ PASS Valid vegan preference [Functional]
Input  : customer_id="C001", preferences=["Vegan"], calorie_limit=2000, session=valid
Expected: success=True, message='Preferences saved successfully'
Actual  : success=True, message='Preferences saved successfully'

B02 ✓ PASS Valid gluten-free preference [ECP Valid]
Input  : customer_id="C001", preferences=["Gluten-free"], calorie_limit=1800, session=valid
Expected: success=True, message='Preferences saved successfully'
Actual  : success=True, message='Preferences saved successfully'

B03 ✓ PASS Calorie limit at lower boundary [BVA Lower]
Input  : customer_id="C001", preferences=["Vegan"], calorie_limit=500, session=valid
Expected: success=True, message='Preferences saved successfully'
Actual  : success=True, message='Preferences saved successfully'

B04 ✓ PASS Calorie limit at upper boundary [BVA Upper]
Input  : customer_id="C001", preferences=["Standard"], calorie_limit=5000, session=valid
Expected: success=True, message='Preferences saved successfully'
Actual  : success=True, message='Preferences saved successfully'

B05 ✓ PASS Calorie limit just below minimum [BVA Lower-1]
Input  : customer_id="C001", preferences=["Vegan"], calorie_limit=499, session=valid
Expected: success=False, error contains 'below minimum'
Actual  : success=False, error='calorie_limit is below minimum allowed (500)'

B06 ✓ PASS Calorie limit just above maximum [BVA Upper+1]
Input  : customer_id="C001", preferences=["Standard"], calorie_limit=5001, session=valid
Expected: success=False, error contains 'exceeds maximum'
Actual  : success=False, error='calorie_limit exceeds maximum allowed (5000)'

B07 ✓ PASS Empty preferences list [Error Handling]
Input  : customer_id="C001", preferences=[], calorie_limit=2000, session=valid
Expected: success=False, error contains 'required'
Actual  : success=False, error='preferences list is required and cannot be empty'
```

B08 ✓ **PASS** Non-numeric calorie limit [ECP Invalid]
Input : customer_id="C001", preferences=["Vegan"], calorie_limit="abc", session=valid
Expected: success=False, error contains 'numeric'
Actual : success=False, error='calorie_limit must be numeric'

B09 ✓ **PASS** Negative calorie limit [BVA Negative]
Input : customer_id="C001", preferences=["Keto"], calorie_limit=-200, session=valid
Expected: success=False, error contains 'negative'
Actual : success=False, error='calorie_limit cannot be negative'

B10 ✓ **PASS** Unknown preference value [ECP Invalid]
Input : customer_id="C001", preferences=["Carnivore_X99"], calorie_limit=2000, session=valid
Expected: success=False, error contains 'Invalid'
Actual : success=False, error='Invalid preference supplied: Carnivore_X99'

B11 ✓ **PASS** Multiple valid preferences [Functional Multi]
Input : customer_id="C001", preferences=["Vegan","Gluten-free"], calorie_limit=1800, session=valid
Expected: success=True, message='Preferences saved successfully'
Actual : success=True, message='Preferences saved successfully'

B12 ✓ **PASS** Missing session token [Error Handling]
Input : customer_id="C001", preferences=["Vegan"], calorie_limit=2000, session=None
Expected: success=False, error contains 'auth'
Actual : success=False, error='authentication failed - session token missing'

B13 ✓ **PASS** Empty customer id [ECP Invalid]
Input : customer_id="", preferences=["Vegan"], calorie_limit=2000, session=valid
Expected: success=False, error contains 'empty'
Actual : success=False, error='customer_id is empty or missing'

B14 ✓ **PASS** Zero calorie limit [BVA Zero]
Input : customer_id="C001", preferences=["Standard"], calorie_limit=0, session=valid
Expected: success=False, error contains 'zero'
Actual : success=False, error='calorie_limit cannot be zero'

B15 ✓ **PASS** SQL-injection style customer id [ECP Invalid]
Input : customer_id="Vegan'; DROP TABLE", preferences=["Vegan"], calorie_limit=2000, session=valid
Expected: success=False, error contains 'invalid'
Actual : success=False, error='customer_id contains invalid characters'

WHITE BOX TESTING (Statement + Branch + Path Coverage)

-
- W01** ✓ **PASS** All statements executed on valid input [Statement Coverage]
Input : customer_id="C001", preferences=["Standard"], calorie_limit=2200, session=valid
Expected: success=True, message='Preferences saved successfully'
Actual : success=True, message='Preferences saved successfully'
- W02** ✓ **PASS** isValid branch = TRUE taken [Branch TRUE]
Input : customer_id="C002", preferences=["Keto"], calorie_limit=1500, session=valid
Expected: success=True, message='Preferences saved successfully'
Actual : success=True, message='Preferences saved successfully'
- W03** ✓ **PASS** isValid branch = FALSE (ValidationException path) [Branch FALSE]
Input : customer_id="", preferences=["Vegan"], calorie_limit=2000, session=valid
Expected: success=False, error contains 'empty'
Actual : success=False, error='customer_id is empty or missing'
- W04** ✓ **PASS** (cal >= MIN) condition = True [Decision Coverage]
Input : customer_id="C003", preferences=["Vegan"], calorie_limit=500, session=valid
Expected: success=True, message='Preferences saved successfully'
Actual : success=True, message='Preferences saved successfully'
- W05** ✓ **PASS** (cal <= MAX) condition = True [Decision Coverage]
Input : customer_id="C003", preferences=["Vegan"], calorie_limit=5000, session=valid
Expected: success=True, message='Preferences saved successfully'
Actual : success=True, message='Preferences saved successfully'
- W06** ✓ **PASS** preferences + calorie conditions both false [Condition Coverage]
Input : customer_id="C004", preferences=[], calorie_limit=-100, session=valid
Expected: success=False, error contains 'required'
Actual : success=False, error='preferences list is required and cannot be empty'
- W07** ✓ **PASS** Full happy path: validate->save->confirm->success [Path Happy]
Input : customer_id="C005", preferences=["Low-sodium"], calorie_limit=2400, session=valid
Expected: success=True, message='Preferences saved successfully'
Actual : success=True, message='Preferences saved successfully'
- W08** ✓ **PASS** Invalid preference: flow stops at validate, no DB write [Path Error]
Input : customer_id="C006", preferences=["Paleo X"], calorie_limit=2000, session=valid
Expected: success=False, error contains 'Invalid'
Actual : success=False, error='Invalid preference supplied: Paleo X'
- W09** ✓ **PASS** Preference loop runs 3 times, all saved [Loop n=3]
Input : customer_id="C007", preferences=["Vegan","Gluten-free","Low-sodium"], calorie_limit=2000, session=valid
Expected: success=True, message='Preferences saved successfully'
Actual : success=True, message='Preferences saved successfully'
- W10** ✓ **PASS** Preference loop runs 0 times, error raised [Loop Zero]
Input : customer_id="C008", preferences=[], calorie_limit=2000, session=valid
Expected: success=False, error contains 'required'
Actual : success=False, error='preferences list is required and cannot be empty'
- W11** ✓ **PASS** DatabaseException caught, transaction rolled back [Exception]
Input : customer_id="C009", preferences=["Vegan"], calorie_limit=2000, session=valid, simulate_db_failure=True
Expected: success=False, error contains 'rolled back'
Actual : success=False, error='database write failed - transaction rolled back'
- W12** ✓ **PASS** isAuthenticated()=False -> AuthException, exits early [Condition Auth]
Input : customer_id="C010", preferences=["Vegan"], calorie_limit=2000, session=None
Expected: success=False, error contains 'auth'
Actual : success=False, error='authentication failed - session token missing'

```

W13a ✓ PASS Enum branch: Vegan [Branch Enum]
Input : customer_id="C011", preferences=["Vegan"], calorie_limit=2000, session=valid
Expected: success=True, message='Preferences saved successfully'
Actual : success=True, message='Preferences saved successfully'

W13b ✓ PASS Enum branch: Keto [Branch Enum]
Input : customer_id="C011", preferences=["Keto"], calorie_limit=2000, session=valid
Expected: success=True, message='Preferences saved successfully'
Actual : success=True, message='Preferences saved successfully'

W13c ✓ PASS Enum branch: Gluten-free [Branch Enum]
Input : customer_id="C011", preferences=["Gluten-free"], calorie_limit=2000, session=valid
Expected: success=True, message='Preferences saved successfully'
Actual : success=True, message='Preferences saved successfully'

W14a ✓ PASS (T,T) -> valid [MC/DC]
Input : customer_id="C012", preferences=["Vegan"], calorie_limit=2000, session=valid
Expected: success=True, message='Preferences saved successfully'
Actual : success=True, message='Preferences saved successfully'

W14b ✓ PASS (T,F) -> error (above maximum) [MC/DC]
Input : customer_id="C012", preferences=["Vegan"], calorie_limit=6000, session=valid
Expected: success=False, error contains 'exceeds maximum'
Actual : success=False, error='calorie_limit exceeds maximum allowed (5000)'

W14c ✓ PASS (F,T) -> error (below minimum) [MC/DC]
Input : customer_id="C012", preferences=["Vegan"], calorie_limit=300, session=valid
Expected: success=False, error contains 'below minimum'
Actual : success=False, error='calorie_limit is below minimum allowed (500)'

W14d ✓ PASS (F,F) -> error (degenerate / non-numeric) [MC/DC]
Input : customer_id="C012", preferences=["Vegan"], calorie_limit="zzz", session=valid
Expected: success=False, error contains 'numeric'
Actual : success=False, error='calorie_limit must be numeric'

```

```

W15a ✓ PASS Path: valid save [Cyclomatic]
Input : customer_id="C013", preferences=["Standard"], calorie_limit=2000, session=valid
Expected: success=True, message='Preferences saved successfully'
Actual : success=True, message='Preferences saved successfully'

W15b ✓ PASS Path: invalid preference type [Cyclomatic]
Input : customer_id="C013", preferences=["BadType"], calorie_limit=2000, session=valid
Expected: success=False, error contains 'Invalid'
Actual : success=False, error='Invalid preference supplied: BadType'

W15c ✓ PASS Path: invalid calorie limit [Cyclomatic]
Input : customer_id="C013", preferences=["Vegan"], calorie_limit=99999, session=valid
Actual : success=False, error='calorie_limit exceeds maximum allowed (5000)'

W15d ✓ PASS Path: database error [Cyclomatic]
Input : customer_id="C013", preferences=["Vegan"], calorie_limit=2000, session=valid, simulate_db_failure=True
Expected: success=False, error contains 'rolled back'
Actual : success=False, error='database write failed - transaction rolled back'

```

=====

TEST SUMMARY

=====

```

Black Box : 15/15 PASSED
White Box : 15/15 PASSED
Total      : 30/30 PASSED
Overall    : PASS

```

```

=====
PS C:\Users\Preet\Downloads\preet_dhote_se> █

```